The spark spread is a proxy for the heat rate (a measure of efficiency) of a specific generating unit or station (the number of Btus needed to make one kilowatt of electricity) multiplied by the cost of energy expressed in dollars per Btu.

For example, if it takes 10,000 British thermal units, or Btus, to make one kilowatt of electricity, the formula can be simplified by multiplying the price per million Btus (MMBtu) by 10 to equate one MMBtu of natural gas to one megawatt hour (Mwh) of electricity. If the heat rate is 6,250, multiply by 6.25; if it is 19,000, multiply by 19.

The accuracy of the spread evaluation is dependent on the market price for power which takes into account operating unit efficiencies. The market price reflects the relationship of the supply and demand for power. Generally, in an over-supplied power market, natural gas may not be the marginal cost energy source. Conversely, in a high-demand period, the cost of natural gas could be lower than the cost of oil, or aggregated, or even imported, power, however, all of the above may be on-line and generating electricity.

Other costs affecting the price of power include those of gas transportation, power transmission, plant operations and maintenance, and fixed costs. Economic dispatch of the next lowest cost generation reflects a utility's ability to cover its costs plus a contribution to overhead based on the wholesale market price of power. The following example of a spark spread calculation shows the basic elements of pricing power using natural gas in the southwestern United States:

Capacity charge per Mwh:	$2.50
Transmission to Palo Verde sub-station/Mwh:	$1.50
Heat rate:	10,000
Natural gas price (El Paso Permian)/MMBtu:	$2.45
Natural gas transmission (burner tip add-on)	$0.18
Total power price per Mwh	$20.30

Using the above values for natural gas at a 10,000 heat rate, the burnertip gas price (energy) equals [($2.45 + $.18) x 10] = $26.30 per Mwh. With the additional $4.00 per Mwh for capacity and transmission charges, the total price to generate power under this scenario is $30.30 per Mwh. (NOTE: This may not represent the market price.)

The spark spread can also reflect the generation costs or anticipated generation costs of power for a specific facility. It can be used as a method of converting millions of Btus to megawatt hours and vice versa. As such, it relates well to the futures contracts for electricity and natural gas.